



LikhaKahoy Woodcraft Shop

Week 12 - Final Project Phase 3 | Insight Report

In this activity, we worked with our woodcraft shop database to create SQL queries that focus on business. The goal was to go beyond just storing data and actually understand what the data is telling us about the business. By doing this, we are able to see patterns and trends that can help in making better decisions.

One of the first things we looked at was the total revenue using the fact_sales table. This gave a quick overview of how the business is performing overall. We also created a query to check monthly revenue trends by grouping data using the date dimension. This makes it easier to see which months have higher or lower sales, which can help in planning promotions or managing stock.

We also identified the top customers based on how much they spend. This is useful because it shows which customers are the most valuable, and the business can focus on keeping them satisfied. Another query focused on best-selling products by looking at which items have the highest total quantity sold. This helps in understanding customer preferences and deciding which products should be restocked more often.

Aside from that, we calculated the average order value to get an idea of how much customers usually spend per purchase. We also looked at revenue by product category to see which categories are doing well and which ones might need improvement.

For the data warehousing part, we transformed the original database into a star schema. The fact_sales table acts as the center, containing important numerical data like quantity and total amount. The dimension tables—customer, product, and date—provide additional details that make analysis easier. This structure makes queries simpler and more efficient, especially when working with large data.

Overall, this activity showed how SQL and data warehousing techniques can be used to turn simple transactional data into meaningful business insights. It also highlighted the importance of organizing data in a way that supports analysis, not just storage. Through this process, I gained a better understanding of how businesses can use data to identify opportunities, improve operations, and make more informed decisions. This activity also showed that even simple queries, when combined properly, can provide valuable insights that are useful for both short-term and long-term planning.